



Image shown may not reflect actual configuration.

## Serial-Ethernet Modbus Gateway

The Serial-Ethernet Modbus Gateway provides connectivity to Serial and Ethernet Devices over Modbus Protocol. The MGate™ MB3170I-T (Cat® PN 657-6662) provides connectivity to a Modbus RTU (remote terminal unit) Master on a serial RS485 bus to Cat Energy Control System (ECS) genset controller on SCADA Modbus Ethernet.

### FEATURES

- Supports Auto Device Routing for easy configuration
- Supports route by IP (internet protocol) address for flexible deployment
- Connects up to 32 Modbus TCP (transmission control protocol) servers
- Connects up to 31 or 62 Modbus RTU/ASCII slaves
- Accessed by up to 32 Modbus TCP clients
- Supports Modbus serial master to Modbus serial slave communications
- Built-in Ethernet cascading for easy wiring
- 10/100BaseTX (RJ45)
- Emergency request tunnels ensure QoS control
- Embedded Modbus traffic monitoring for easy troubleshooting
- Serial port with 2 kV isolation protection
- -40 to 75°C wide operating temperature
- Supports redundant dual DC power inputs and 1 relay output
- Priority control feature addresses lag and response time issue associated with large complex Modbus networks. It provides prioritization for urgent commands, allowing users to force certain commands to get an immediate response. Different methods and settings are available to define which commands receive priority depending on network system requirements.

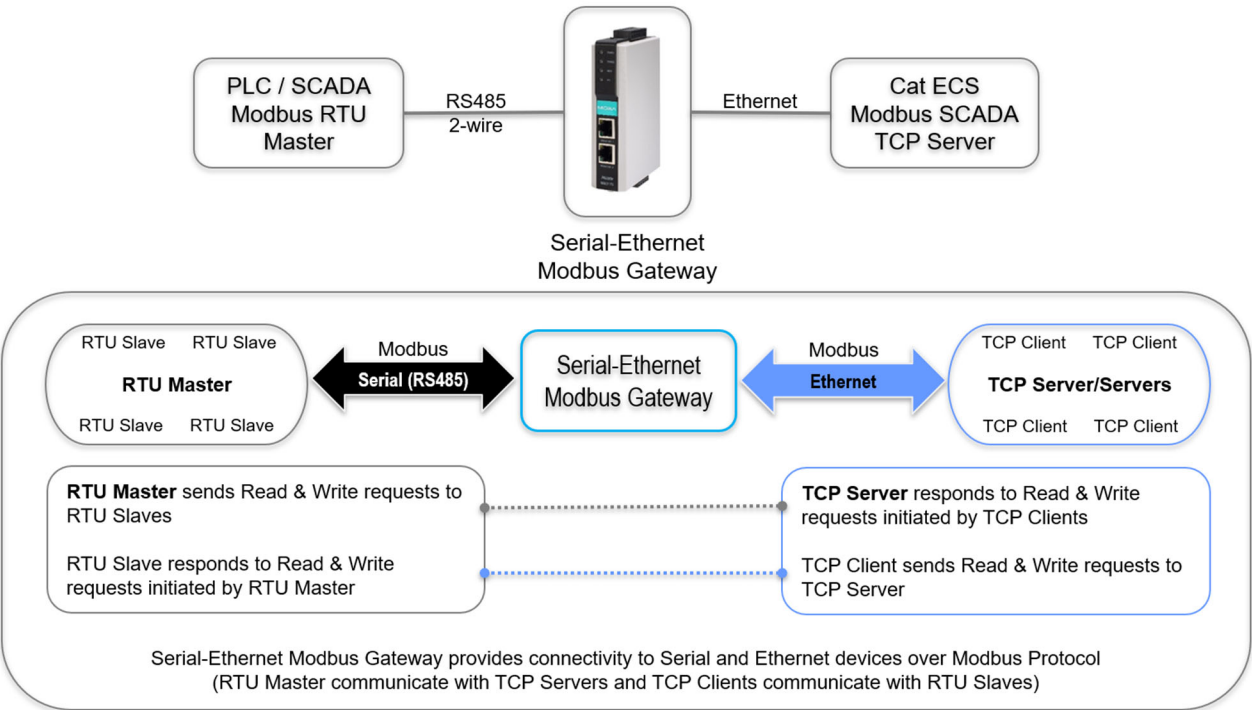
### INTRODUCTION

The MGate™ MB3170I-T (Cat PN 657-6662) is a rugged DIN-rail mountable serial-Ethernet Modbus gateway device.

The MB3170I-T is a 1-port serial Modbus gateway which converts between Modbus TCP, ASCII, and RTU communications protocols.

It provides both serial-to-Ethernet communication and serial (master) to serial (slave) communications. In addition, the gateway supports simultaneously connecting serial and Ethernet masters with serial Modbus devices. Serial-Ethernet Modbus Gateway can be accessed by up to 32 TCP master/clients or connect to up to 32 TCP slave/servers. Serial-port routing by ID mapping allows access for up to 32 TCP clients/masters. A featured priority control function allows urgent commands to obtain an immediate response.

## Serial-Ethernet Modbus Gateway



### Technical Specification

| Ethernet Interface                    |                                                         |
|---------------------------------------|---------------------------------------------------------|
| 10/100BaseT(X) Ports (RJ45 connector) | 2 (1 IP, Ethernet cascade)<br>Auto MDI/MDI-X connection |
| Magnetic Isolation Protection         | 1.5 kV (built-in)                                       |

| Ethernet Software     |                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| Industrial Protocols  | Modbus TCP Client (Master)<br>Modbus TCP Server (Slave)                                                                   |
| Configuration Options | Web Console (HTTP/HTTPS)<br>Device Search Utility (DSU)<br>MGate™ Manager<br>MCC Tool<br>Telnet Console                   |
| Management            | ARP<br>DHCP Client<br>DNS<br>HTTP<br>HTTPS<br>SMTP<br>SNMP Trap<br>SNMPv1/v2c/v3<br>TCP/IP<br>Telnet<br>UDP<br>NTP Client |
| MIB                   | RFC1213, RFC1317                                                                                                          |
| Time Management       | NTP Client                                                                                                                |

## Technical Specification (continued)

| Security Functions |                                        |
|--------------------|----------------------------------------|
| Authentication     | Local database                         |
| Encryption         | HTTPS<br>AES-128<br>AES-256<br>SHA-256 |
| Security Protocols | SNMPv3<br>HTTPS (TLS 1.2)              |

| Serial Interface                  |                                                    |
|-----------------------------------|----------------------------------------------------|
| No. of Ports                      | 1                                                  |
| Connector                         | DB9 male for RS-232, Terminal block for RS-422/485 |
| Serial Standards                  | RS-232/422/485 (software selectable)               |
| Baud rate                         | 50 bps to 921.6 kbps                               |
| Data Bits                         | 7, 8                                               |
| Parity                            | None<br>Even<br>Odd<br>Space<br>Mark               |
| Stop Bits                         | 1, 2                                               |
| Flow Control                      | DTR/DSR<br>RTS Toggle (RS-232 only)<br>RTS/CTS     |
| Pull High/Low Resistor for RS-485 | 1 kilo-ohm, 150 kilo-ohms                          |
| Terminator for RS-485             | 120 ohms                                           |
| Isolation                         | 2 kV                                               |
| RS-485 Data Direction Control     | ADDC (automatic data direction control)            |

| Serial Signals |                                        |
|----------------|----------------------------------------|
| RS-232         | TxD, RxD, RTS, CTS, DTR, DSR, DCD, GND |
| RS-422         | Tx+, Tx-, Rx+, Rx-, GND                |
| RS-485-2w      | Data+, Data-, GND                      |
| RS-485-4w      | Tx+, Tx-, Rx+, Rx-, GND                |

| Serial Software Features |                                                   |
|--------------------------|---------------------------------------------------|
| Industrial Protocols     | Modbus RTU/ASCII Master<br>Modbus RTU/ASCII Slave |

| Modbus (Transparent)           |    |
|--------------------------------|----|
| Max. No. of Client Connections | 32 |
| Max. No. of Server Connections | 32 |

## Technical Specification (continued)

| Power Parameters |                      |
|------------------|----------------------|
| Input Voltage    | 12 to 48 VDC         |
| Input Current    | 435 mA @ 12 VDC      |
| Power Connector  | 7-pin terminal block |

| Relays                 |                              |
|------------------------|------------------------------|
| Contact Current Rating | Resistive load: 1 A @ 30 VDC |

| Physical Characteristics  |                                              |
|---------------------------|----------------------------------------------|
| Housing                   | Plastic                                      |
| IP Rating                 | IP30                                         |
| Dimensions (with ears)    | 29 x 89.2 x 124.5 mm (1.14 x 3.51 x 4.90 in) |
| Dimensions (without ears) | 29 x 89.2 x 118.5 mm (1.14 x 3.51 x 4.67 in) |
| Weight                    | MGate™ MB3170 models: 360 g (0.79 lb)        |

| Environmental Limits                   |                            |
|----------------------------------------|----------------------------|
| Operating Temperature                  | -40 to 75°C (-40 to 167°F) |
| Storage Temperature (package included) | -40 to 85°C (-40 to 185°F) |
| Ambient Relative Humidity              | 5 to 95% (non-condensing)  |

| Standards and Certifications |                                                                                                                                                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMC                          | EN 55032/35                                                                                                                                                                                                                                    |
| EMI                          | CISPR 32, FCC Part 15B Class A                                                                                                                                                                                                                 |
| EMS                          | IEC 61000-4-2 ESD: Contact: 6 kV; Air: 8 kV<br>IEC 61000-4-3 RS: 80 MHz to 1 GHz: 10 V/m<br>IEC 61000-4-4 EFT: Power: 4 kV; Signal: 2 kV<br>IEC 61000-4-5 Surge: Power: 2 kV<br>IEC 61000-4-6 CS: 10 V<br>IEC 61000-4-8 PFMF<br>IEC 61000-4-11 |
| Hazardous Locations          | ATEX<br>Class I Division 2<br>IECEX                                                                                                                                                                                                            |
| Maritime                     | DNV                                                                                                                                                                                                                                            |
| Freefall                     | IEC 60068-2-32                                                                                                                                                                                                                                 |
| Shock                        | IEC 60068-2-27                                                                                                                                                                                                                                 |
| Vibration                    | IEC 60068-2-6<br>IEC 60068-2-64                                                                                                                                                                                                                |
| Safety                       | UL 60950-1<br>IEC 60950-1<br>EN 62368-1<br>UL 508                                                                                                                                                                                              |

# Serial-Ethernet Modbus Gateway



## Dimensions

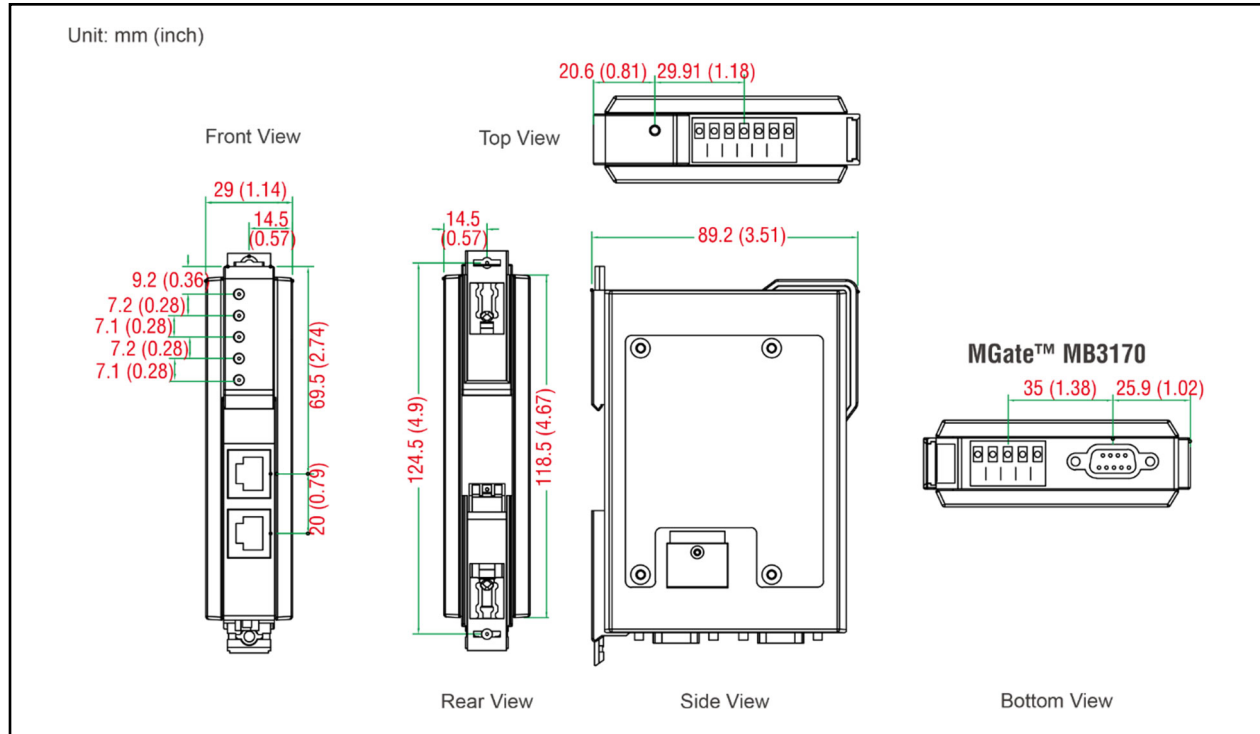


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## Worldwide Product Support

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